

Firefox DevTools

Kameswari Chebrolu

Department of CSE, IIT
Bombay



<https://starecat.com/content/wp-content/uploads/firefox-logo-evolution-look-how-they-massacred-my-boy.jpg>

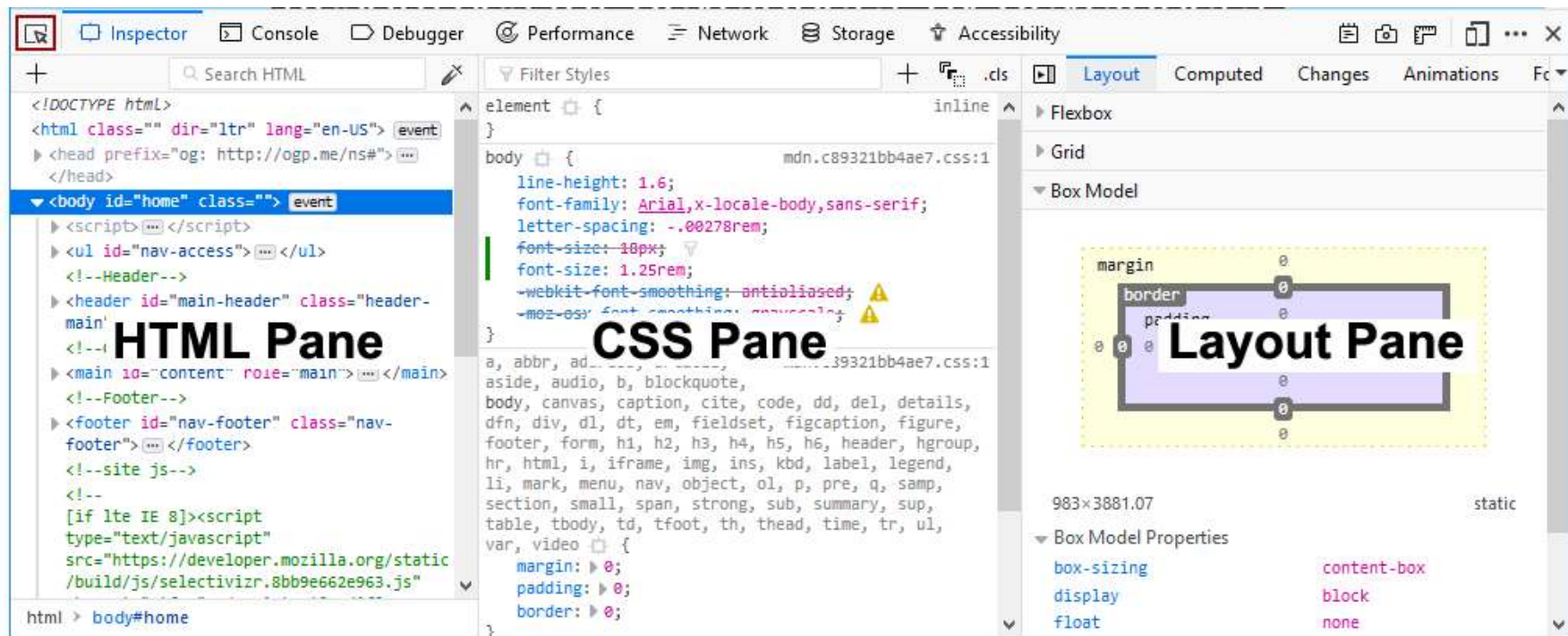
Introduction

- A set of web developer tools built into Firefox
- Can use them to examine, edit, and debug HTML, CSS, and JavaScript
- How to open?
 - More Tools > Web Developer > Web Developer Tools
 - Keyboard shortcut Ctrl + Shift + I or F12 on Windows and Linux, or Cmd + Opt + I on macOS.
- How to dock?
 - Click on 3 dots (ellipsis)
- How to close?
 - Click on the cross

Outline

- Page Inspector
- Web Console
- Network
- Storage

Page Inspector



- Page Inspector helps examine and modify the HTML and CSS of a page
- Important things to know
 - Select element button 
 - Examine and edit HTML 

Web Console



- Logs information associated with a web page: network requests, JavaScript, CSS, security errors and warnings
- Error, warning and informational messages explicitly logged by JavaScript code running in the page'
- Enables you to interact with a web page by executing JavaScript expressions in the context of the page
- Important things to know
 - Settings
 - Clear Log
 - Javascript execution

Network

The screenshot shows the Network panel in Chrome DevTools. Red arrows point to various UI elements: 'Clear log' (trash icon), 'Filter by URL' (filter icon), 'Tools' (three arrows icon), 'Filter by type' (dropdown menu), 'Cache control' (checkbox), 'Throttling' (dropdown), and 'Action menu' (gear icon). The table below lists network requests with columns for Status, Method, Domain, File, Initiator, Type, Transfer Size, and Size. The first four rows are visible, all showing 200 status and CSS files from developer.mozilla.org.

Status	M...	Domain	File	Initiator	T...	Transfer...	Siz
200	G...	developer.mozilla.org	home_featured.2b4900e0000a.css	stylesheet	C...	cached	2.1
200	G...	developer.mozilla.org	banners.92084d28db45.css	stylesheet	c...	cached	1.9
200	G...	developer.mozilla.org	mdn-subscriptions.dbb82432d74c.css	stylesheet	c...	cached	86
200	G...	developer.mozilla.org	print.ca192a439743.css	stylesheet	c...	cached	32



- Shows all the HTTP requests Firefox makes, how long each request takes, and details of each request
- Things to Know
 - Toolbar
 - Network Request List
 - Network Request Details
 - Edit a request and resend

Storage

The screenshot shows the Chrome DevTools Storage panel. The left sidebar (Storage Tree) lists Cache Storage, Cookies, Local Storage, and Session Storage. The Cookies section is expanded, showing a list of cookies. The _ga cookie is selected, and its details are shown in the right sidebar (Sidebar). The details include CreationTime, Domain, Expires, HostOnly, HttpOnly, LastAccessed, Path, Secure, and sameSite.

Storage Tree

- Cache Storage
 - https://developer.mozilla.org
- Cookies
 - https://developer.mozilla.org
- Local Storage
- Session Storage

Table Widget

Name	Domain	Path	Expires on	Last access
_ga	.mozilla.org	/	Tue, 23 Feb 2021 1...	Sun, 23 Ju
csrftoken	.developer...	/	Mon, 08 Jun 2020 ...	Sun, 23 Ju
csrftoken	developer...	/	Tue, 26 Mar 2020 0...	Sun, 23 Ju
dwf_sg_t...	.developer...	/	Tue, 26 Mar 2020 18...	Sun, 23 Ju
lux_uid	developer...	/	Sun, 23 Jun 2019 1...	Sun, 23 Ju

Sidebar

Data

- _ga: "GA1.2.1812154749.1526560462"
- CreationTime: "Thu, 17 May 2018 12:34:22 GMT"
- Domain: ".mozilla.org"
- Expires: "Tue, 23 Feb 2021 12:00:00 GMT"
- HostOnly: false
- HttpOnly: false
- LastAccessed: "Sun, 23 Jun 2019 18:15:54 GMT"
- Path: "/"
- Secure: false
- sameSite: "Unset"

- Can inspect various types of storage that a web page can use
 - Cache Storage — any DOM caches created using the Cache API
 - Cookies — All the cookies created by the page or any iframes inside of the page
 - IndexedDB — All IndexedDB databases created by the page or any iframes inside the page
 - Local Storage — All local storage items created by the page or any iframes inside the page
 - Session Storage — All session storage items created by the page or any iframes inside the page
- Things to Know (Cookies)
 - Column Details
 - Search
 - Add, Edit and Delete

Persistent

non-persistent

References

- https://firefox-source-docs.mozilla.org/devtools-user/page_inspector/index.html
- https://firefox-source-docs.mozilla.org/devtools-user/web_console/index.html
- https://firefox-source-docs.mozilla.org/devtools-user/network_monitor/index.html
-  https://firefox-source-docs.mozilla.org/devtools-user/storage_inspector/index.html